AMS 526: Numerical Linear Algebra for Computational and Data Sciences

Lecture 1: Course Overview; Matrix Multiplication

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Outline

1. **Course Overview**
2. Matrix Notation and Basic Operations (MC §1.1)
3. Range, Null Space and Rank (NLA §1)

Course Description

- What is numerical linear algebra?
  - Solving linear algebra problems using efficient algorithms on computers.
- Topics covered include:
  - Direct and iterative methods for solving simultaneous linear equations.
  - Least squares problems.
  - Computation of eigenvalues and eigenvectors.
  - Singular value decomposition.
- Required textbooks:
  - **[NLA]** L. N. Trefethen and D. Bau III, *Numerical Linear Algebra*, SIAM, 1997.
  - **[MC]** G. H. Golub and C. F. Van Loan, *Matrix Computations*, 4th Edition, Johns Hopkins University Press, 2012. ISBN 978-1421407944.
- Additional supplementary material will be provided as needed.
- Course webpage: `http://www.ams.sunysb.edu/~jiao/teaching/ams526`

Prerequisite

- You **MUST NOT** take this as your first course in linear algebra, or you **will** get lost.
- Prerequisite:
  - AMS 510 (linear algebra portion) or equivalent undergraduate-level linear algebra course. Familiarity with the following concepts is assumed: Vector spaces, Gaussian elimination, Gram–Schmidt orthogonalization, and eigenvalues/eigenvectors.
  - AMS 595 (for students without programming experience).
- To review fundamental concepts of linear algebra, refer to textbooks such as:
  - Gilbert Strang, *Linear Algebra and Its Applications*, 4th Edition, Brooks Cole, 2006.

Why Learn Numerical Linear Algebra?

- Foundation of scientific computations and data sciences.
- Many problems ultimately reduce to linear algebra concepts or algorithms, either analytical or computational. For example:
  - Computational and Applied Mathematics:
    - Finite-element analysis involves solving large-scale linear systems.
    - Data compression often utilizes singular value decomposition (SVD).
  - Search Engine and Recommendation Systems:
    - Google's PageRank solves an approximate eigenvalue problem.
    - Matrix factorization techniques (e.g., collaborative filtering) are foundational for recommendation systems (Netflix, Amazon).
  - Machine Learning and Deep Learning:
    - Support Vector Machines (SVM) rely on solving linear equations and quadratic programming problems.
    - Principal Component Analysis is rooted in eigenvalue problems and SVD.
    - Training algorithms often involve optimization problems heavily relying on numerical linear algebra.
- Focus: Fundamental concepts, efficiency and stability of algorithms, and programming.

Course Outline

- **Fundamentals**: Matrix notation and basic operations; vector spaces; algorithmic considerations; norms and condition numbers; decomposition of matrices.
- **Linear systems**: Triangular systems; Gaussian elimination; accuracy and stability; Cholesky factorization.
- **QR factorization and least squares**: Gram–Schmidt orthogonalization; QR factorization with Householder reflection; updating QR factorization with Givens rotation; stability; least squares problems; rank-revealing QR; SVD and low-rank approximations.
- **Eigenvalue problems**: Eigenvalues and invariant spaces; classical eigenvalue methods; QR algorithms; two-stage methods; Arnoldi and Lanczos iterations.
- **Iterative methods for linear systems**: Basic iterative methods; conjugate gradient methods; minimal residual style methods; bi-Lanczos iterations; preconditioners.
- **Special topics**: Multigrid methods; under-determined systems, etc., if time permits.

Course Policy

- Assignments (written and programming):
  - Posted on Brightspace; typically due 2 weeks after assigned.
  - **Late Policy**: 24h grace period (-10%), 24–48h (-20%).
  - Discuss concepts okay, but write answers independently.
  - Do NOT copy solutions. Do NOT share solutions.
  - **Generative AI**: OK for grammar/debugging (no credit needed). Must cite if used for substantial content. Do not use for problem solving.
- Exams:
  - Closed-book.
  - **Single-sided**, one-page cheat sheet allowed.
- Grading:
  - Assignments: 20%
  - Two midterm exams: 40% (20% each)
  - Final exam: 30%
  - In-class exercises/quizzes: 10% (Calculated via weighted average of exams & attendance).

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Matrices and Vectors

- Denote vector space of all m-by-n real matrices by $\mathbb{R}^{m\times n}$.

$$A \in \mathbb{R}^{m\times n} \iff A = (a_{ij}) = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

- Denote vector space of all real n-vectors by $\mathbb{R}^n$, or $\mathbb{R}^{n\times 1}$.

$$x \in \mathbb{R}^n \iff x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

- Transposition ($\mathbb{R}^{m\times n} \to \mathbb{R}^{n\times m}$): $C = A^\top \implies c_{ij} = a_{ji}$.
- Row vectors are transpose of column vectors and are in $\mathbb{R}^{1\times n}$.

Matrix Operations

- *Addition* and *subtraction* ($\mathbb{R}^{m\times n} \times \mathbb{R}^{m\times n} \to \mathbb{R}^{m\times n}$): $C = A \pm B \implies c_{ij} = a_{ij} \pm b_{ij}$.
- *Scalar-matrix multiplication* or *scaling* ($\mathbb{R} \times \mathbb{R}^{m\times n} \to \mathbb{R}^{m\times n}$): $C = \alpha A \implies c_{ij} = \alpha a_{ij}$.
- *Matrix-matrix multiplication/product* ($\mathbb{R}^{m\times p} \times \mathbb{R}^{p\times n} \to \mathbb{R}^{m\times n}$):
  $C = AB \implies c_{ij} = \sum_{k=1}^{p} a_{ik} b_{kj}$ (denoted by $A * B$ in MATLAB).
- Each operation also applies to vectors. In particular:
  - *Inner product* is row vector times column vector, i.e., $c = x^{\mathsf{T}} y$ (called *dot product* in vector calculus and denoted as $x \cdot y$).
  - *Outer product* is column vector times row vector, i.e., $C = xy^{\mathsf{T}}$ (a special case of Kronecker product, denoted as $x \otimes y$).
- *Element-wise multiplication* and *division* ($\mathbb{R}^{m\times n} \times \mathbb{R}^{m\times n} \to \mathbb{R}^{m\times n}$):
  - $C = A.*B \implies c_{ij} = a_{ij} b_{ij}$.
  - $C = A./B \implies c_{ij} = a_{ij}/b_{ij}$, where $b_{ij} \neq 0$.
- *Matrix inversion* (A^{-1}) and *division* (A/B and $A\backslash B$) to be defined later.

Notation of Matrices and Vectors

- Matrix notation:
  - Capital letters (e.g., A, B, Δ) for matrices.
  - Corresponding lower case with subscript ij (e.g., a_{ij}, b_{ij}, δ_{ij}) for the (i, j) entry; sometimes notation $[A]_{ij}$ or $A(i, j)$ is used.
- Vector notation:
  - Lowercase letters (e.g., x, y) for vectors.
  - Corresponding lower case with subscript i for ith entry (e.g., x_i, y_i).
- Lowercase letters for scalars (e.g., c, s, α, β).
- Some books suggest using boldface lowercase (e.g., x) for vectors, regular lowercase (e.g., c) for scalars, and boldface uppercase (e.g., A) for matrices.
- A matrix is a collection of column vectors or row vectors:

$$A \in \mathbb{R}^{m\times n} \iff A = [c_1|c_2|\dots|c_n], \quad c_k \in \mathbb{R}^m$$

$$A \in \mathbb{R}^{m\times n} \iff A = \begin{bmatrix} r_1^\top \\ \vdots \\ r_m^\top \end{bmatrix}, \quad r_k \in \mathbb{R}^n$$

Complex Matrices

- Occasionally, complex matrices are involved.
- Vector space of m-by-n complex matrices is designated by $\mathbb{C}^{m\times n}$.
  - Scaling, addition, multiplication of complex matrices correspond exactly to the real case.
  - If $A = B + iC \in \mathbb{C}^{m\times n}$, then $\mathrm{Re}(A) = B$, $\mathrm{Im}(A) = C$, and the *conjugate* of A is $\overline{A} = (\overline{a}_{ij})$.
  - *Conjugate transpose* is defined as $A^{\mathrm{H}} = B^{\mathrm{T}} - iC^{\mathrm{T}}$, or

$$G = A^{\mathrm{H}} \implies g_{ij} = \overline{a}_{ji}$$

    (also called *adjoint* and denoted by A^*; $(AB)^* = B^*A^*$).
- Vector space of complex n-vectors is designated by $\mathbb{C}^n$.
  - Inner product of complex n-vectors x and y is $s = x^{\mathrm{H}}y$.
  - Outer product of complex n-vectors x and y is $S = xy^{\mathrm{H}}$.
- We will primarily focus on real matrices.

Matrix-Vector Product

- Matrix-vector product $b = Ax$ is a special case of matrix-matrix product:

$$b_i = \sum_{j=1}^{n} a_{ij} x_j.$$

- For $A \in \mathbb{R}^{m \times n}$, Ax is a mapping $x \longmapsto Ax$ from $\mathbb{R}^n$ to $\mathbb{R}^m$.
- This map is *linear*, which means that for any $x, y \in \mathbb{R}^n$ and any $\alpha \in \mathbb{R}$:

$$A(x + y) = Ax + Ay$$
$$A(\alpha x) = \alpha Ax$$

Linear Combination

- Let a_j denote the jth column of matrix A.
  - An alternative notation is colon notation: $A(:, j)$ or $a_{:,j}$.
  - Use $A(i, :)$ or $a_{i,:}$ to denote the ith row of A.
- b is a *linear combination* of column vectors of A, i.e.,

$$b = Ax = \sum_{j=1}^{n} x_j a_j = x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1m} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

- In summary, there are two different views of matrix-vector products:
  1. Scalar operations: $b_i = \sum_{j=1}^{n} a_{ij} x_j$ (A acts on x to produce b).
  2. Vector operations: $b = \sum_{j=1}^{n} x_j a_j$ (x acts on A to produce b).

Matrix-Matrix Multiplication

- Compute $C = AB$, where $A \in \mathbb{R}^{m \times r}$, $B \in \mathbb{R}^{r \times n}$, and $C \in \mathbb{R}^{m \times n}$.
- Element-wise, each entry of C is

$$c_{ij} = \sum_{k=1}^{r} a_{ik} b_{kj}.$$

- Column-wise, each column of C is

$$c_j = Ab_j = \sum_{k=1}^{r} b_{kj} a_k;$$

in other words, the jth column of C is A times the jth column of B.

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Terminology of Vector Space

- *Vector space* is closed under addition and scalar multiplication, with the zero vector as a member.
- Vector space *spanned* by a set of vectors $\{a_j\}$ in $\mathbb{R}^m$ is

$$\text{span}\{a_1, \dots, a_n\} = \left\{ \sum_{j=1}^{n} \beta_j a_j \mid \beta_j \in \mathbb{R} \right\}.$$

  - The space spanned by n-vectors is a *subspace* of $\mathbb{R}^m$.
  - If $\{a_1, \dots, a_n\}$ is *linearly independent*, then the a_j are the *basis* of $S = \text{span}\{a_1, \dots, a_n\}$, *dimension* of S is $\dim(S) = n$, and each $b \in S$ is a unique linear combination of the a_j.
- If S_1 and S_2 are two subspaces, then $S_1 \cap S_2$ is a subspace, as is $S_1 + S_2 = \{b_1 + b_2 \mid b_1 \in S_1, b_2 \in S_2\}$.
  - Note: $S_1 + S_2$ is different from $S_1 \cup S_2$; the latter may not be a subspace.
- Two subspaces S_1 and S_2 of $\mathbb{R}^m$ are *complementary subspaces* of each other if $S_1 + S_2 = \mathbb{R}^m$ and $S_1 \cap S_2 = \{0\}$.
  - In other words, $\dim(S_1) + \dim(S_2) = m$ and $S_1 \cap S_2 = \{0\}$.

Range and Null Space

Definition

The *range* of a matrix A, written as range(A) or ran(A), is the set of vectors that can be expressed as Ax for some x: $\{y \in \mathbb{R}^m \mid y = Ax \text{ for some } x \in \mathbb{R}^n\}$.

Theorem

range(A) is the space spanned by the columns of A.

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The entries of $x \in \text{null}(A)$ give the coefficients of $\sum x_i a_i = 0$.
Note: The null space of A is in general **not** a complementary subspace of range(A).

Relationship Between Null and Range Space

- For real matrices $A \in \mathbb{R}^{m \times n}$:
  - $\text{null}(A)$ and $\text{range}(A^{\mathsf{T}})$ are complementary subspaces.
  - For symmetric matrices ($A = A^{\mathsf{T}}$), $\text{null}(A)$ and $\text{range}(A)$ are complementary subspaces.
- For complex matrices $A \in \mathbb{C}^{m \times n}$:
  - $\text{null}(A)$ and $\text{range}(A^{\mathsf{H}})$ are complementary subspaces.
  - For *Hermitian* matrices ($A = A^{\mathsf{H}}$), $\text{null}(A)$ and $\text{range}(A)$ are complementary subspaces.

Rank

Definition

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Answer: No. We will give a proof in later lectures.

- Therefore, we simply refer to it as the *rank* of a matrix.

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Question: Given $A \in \mathbb{R}^{m \times n}$, what is $\dim(\text{null}(A)) + \text{rank}(A)$ equal to?
Answer: n.

- A real matrix A is of rank-1 if it can be written as $A = uv^\top$, where u and v are nonzero vectors.

Full Rank

Definition

A matrix has *full rank* if it has the maximal possible rank, i.e., $\min\{m, n\}$.

Otherwise, it is called *rank deficient*.

Theorem

A matrix $A \in \mathbb{R}^{m \times n}$ with $m \geq n$ has full rank if and only if it maps no two distinct vectors to the same vector.

Proof

("only if") Column vectors of A form a basis of range(A), so every $b \in$ range(A) has a unique linear expansion in columns of A.
("if") If A does not have full rank, then its column vectors are linearly dependent, so its vectors do not have a unique linear combination.

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Full Rank vs. Non-singularity

- If $A \in \mathbb{R}^{n\times n}$ and $AX = I$, then X is the *inverse* of A, denoted by A^{-1}.
  - $(AB)^{-1} = B^{-1}A^{-1}$.
  - $(A^{-1})^{\top} = (A^{\top})^{-1} = A^{-\top}$.
  - Sherman–Morrison–Woodbury formula for $U, V \in \mathbb{R}^{n\times k}$:

$$(A + UV^{\top})^{-1} = A^{-1} - A^{-1}U(I + V^{\top}A^{-1}U)^{-1}V^{\top}A^{-1}.$$

- If A^{-1} exists, A is *nonsingular*. A is square and has full rank.
- If A is nonsingular, the linear system $Ax = b$ results in $x = A^{-1}b$, and it is the inverse problem of matrix-vector multiplication.
- If $A \in \mathbb{R}^{m\times n}$ where $n \neq m$ and A has full rank, then what is the inverse problem of matrix-vector multiplication?
- If A is rank deficient, then what is the inverse problem of matrix-vector multiplication?
- We will need progressively more advanced linear algebra concepts to answer these questions in the later part of this semester.